



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2024-25(Even Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

END SEMESTER

Semester: VI

Programme: B. Tech Information Technology

Course: Natural Language Processing

Time: 3 Hours

Max Marks: 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT33608	Course Outcome
CO1	Classify the ability to preprocess text data using NLP techniques.
CO2	Demonstrate syntactic parsing methods, including context-free grammars and dependency parsing
CO3	Analyze semantic structures and apply information retrieval techniques to extract meaningful insights from text.
CO4	Interpret language models and perform text classification using algorithms in NLP.
CO5	Evaluate real-world applications of Natural Language Processing

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	What is tokenization?	CO1	2	2
b.	Define morphology.	CO2	2	2
c.	What is semantic role?	CO3	2	2
d.	Define SVM.	CO4	3	2
e.	What are chatbots?	CO5	2	2
Group B				

Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain text preprocessing steps and their importance.	CO1	2	5
b.	Discuss challenges in NLP (ambiguity, context, sarcasm)	CO1	2	5
c.	Explain FastText and BERT embeddings (difference).	CO1	2	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Compare rule-based, statistical and transformer-based POS tagging.	CO2	2	5
b.	Explain chunking and its role in NLP.	CO2	3	5
c.	Describe ambiguity in syntax with examples.	CO2	3	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain semantic networks and their applications.	CO3	2	5
b.	Discuss relevance ranking in Information Retrieval.	CO3	3	5
c.	Explain WSD using machine learning approach.	CO3	4	5
Que 5.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain unigram vs bigram vs trigram with limitations.	CO4	2	5
b.	Compare Naive Bayes and SVM classifiers.	CO4	3	5
c.	Explain LSTM architecture and its use in NLP.	CO4	4	5
Que 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain sentiment analysis pipeline.	CO5	2	5
b.	Describe working of dialogue systems.	CO5	3	5
c.	Explain transformer networks (basic architecture).	CO5	4	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create